Kitchen Sink

Every Feature in One File

March 25, 2026

Abstract

This document exercises every supported feature of Calepin in a single file.

Keywords test, features

0.1 Markdown basics

This section tests standard CommonMark and GFM features.

0.1.1 Inline formatting

Italic, **bold**, ***bold italic***, ~~strikethrough~~, and `inline code`.

0.1.2 Lists

Ordered:

1. First item
2. Second item
3. Third item

Unordered:

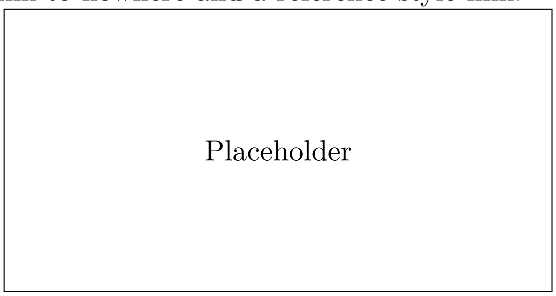
- Alpha
- Beta
- Gamma

0.1.3 Blockquote

> This is a blockquote with **bold** and *italic* text.

0.1.4 Links and images

A link to nowhere and a reference-style link.



0.1.5 Footnotes

Here is a sentence with a footnote[1].

0.1.6 Horizontal rule

0.1.7 Table

Name	Score	Grade
Alice	95	A
Bob	87	B
Carol	92	A

0.2 Code chunks

0.2.1 R chunks

```
x <- 1:10
mean(x)
```

```
> [1] 5.5
```

[1] This is the footnote content.

Echo fenced

```
```{r, chunk-2}
#| echo: fenced
summary(mtcars$mpg)
```
```

```
>    Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
>   10.40   15.43   19.20   20.09   22.80   33.90 
```

Figure from R

```
plot(1:10, (1:10)^2, pch = 19, col = "steelblue",
     xlab = "x", ylab = "x squared")
```

See Figure 1 for the scatter plot.

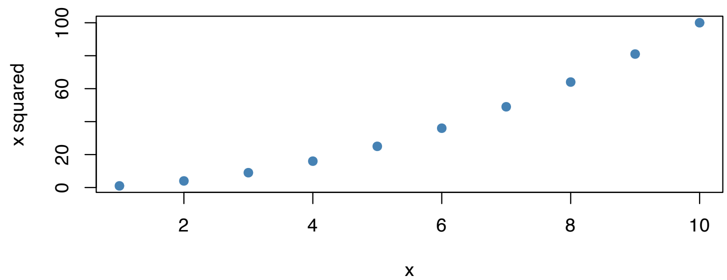


Figure 1: Scatter plot of x vs x squared

Echo false

Eval false

```r
stop("this would error if evaluated")
```

Include false (silent execution)

Results asis

```r
cat("**This is bold from asis output.**\n")
```

This is bold from asis output.

Warning and message control

```
warning("suppressed warning")
message("suppressed message")
cat("Only this output appears.\n")
```

```
> Only this output appears.
```

0.2.2 Python chunks

```
import math
print(f"Pi is approximately {math.pi:.4f}")
```

```
> Pi is approximately 3.1416
```

```python
squares = [x**2 for x in range(1, 6)]
print(squares)
```

```
> [1, 4, 9, 16, 25]
```

0.2.3 Shell chunks

```bash
echo "Hello from shell"
```

```
> Hello from shell
```

0.2.4 Inline code

The mean of 1 to 10 is 5.5. Pi is 3.14. The secret value is 42.

0.3 Math

0.3.1 Inline math

The Pythagorean theorem states $a^2 + b^2 = c^2$.

A literal dollar sign costs $5.

0.3.2 Display math

$$\int_0^1 x^2 \, dx = \frac{1}{3}$$

0.3.3 Labeled equation

$$E = mc^2 \tag{1}$$

0.3.4 Another labeled equation

$$\sum_{i=1}^{n} i = \frac{n(n+1)}{2} \tag{2}$$

See Equation 1 and Equation 2.

0.4 Cross-references

We can reference sections (Section 1), figures (Figure 1), tables (Table 1), theorems (Theorem 1), definitions (Definition 1), and code listings (Listing 1).

Bracketed form: [Figure 1].

Suppress type: 1. Grouped: [Equation 1; Equation 2].

0.5 Callouts

Note

0.5.1 A note

This is a **note** callout with a custom title.

Tip

A tip callout with the default title.

Warning

0.5.2 Watch out

A warning callout.

Important

An important callout with the default title.

Caution

0.5.3 Collapsible caution

This caution callout starts collapsed.

0.6 Theorems

Theorem 1.

0.6.1 Fermat's Last Theorem

For $n > 2$*, there are no positive integers* a*,* b*,* c *such that* $a^n + b^n = c^n$*.*

Proof. The proof was completed by Andrew Wiles in 1995 and is too long to include here.

□

Definition 1.

0.6.2 Group

A group is a set G with a binary operation satisfying closure, associativity, identity, and invertibility.

Lemma 1. *Every finite group has an element of order dividing the group order.*

Corollary 1. *Every group of prime order is cyclic.*

Example 1. The integers under addition form a group.

Remark 1. Groups are fundamental to abstract algebra.

By Theorem 1 and Definition 1, we see the connection. Also see Lemma 1, Corollary 1, Example 1, and Remark 1.

0.7 Tabsets

0.7.1 Tab A

Content of the first tab.

0.7.2 Tab B

Content of the second tab with some math: $e^{i\pi} + 1 = 0$.

0.7.3 Tab C

```
cat("Code in a tab\n")
```

```
> Code in a tab
```

0.8 Layouts

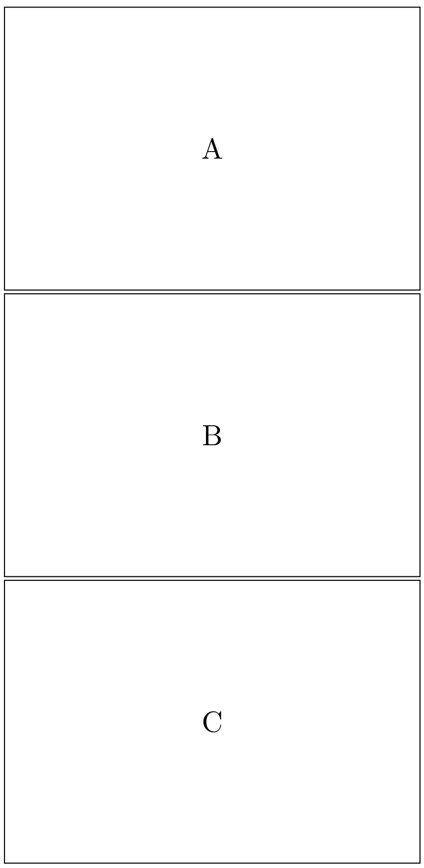
0.8.1 Two-column layout

Left column content. This paragraph appears on the left side of a two-column layout.
Right column content. This paragraph appears on the right side.

Table 1: Number of known moons for selected planets.

Planet	Moons
Earth	1
Mars	2
Jupiter	95

0.8.2 Custom grid layout



0.9 Figure divs

See Figure 1.

0.10 Table divs

See Table 1.

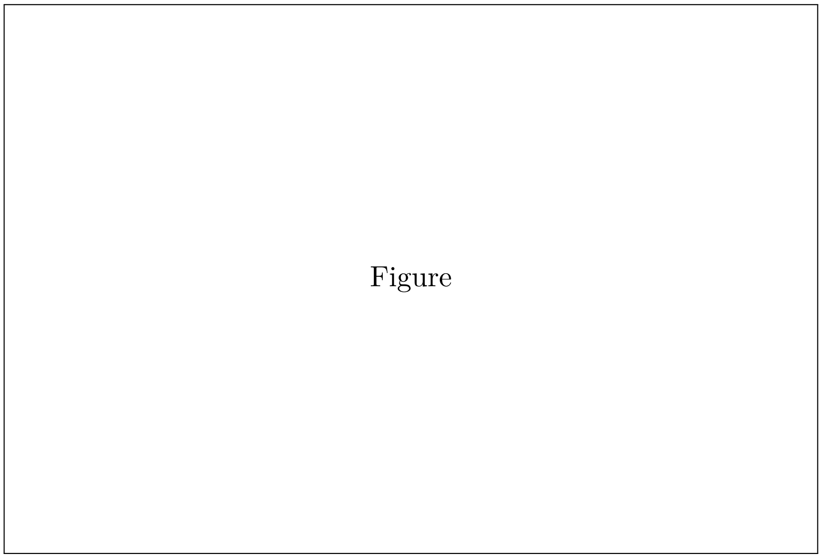


Figure 2: A figure div with a caption.

0.11 Code listings

```r
square <- function(x) x^2
square(7)
```

```
> [1] 49
```

See Listing 1.

0.12 Content visibility

This paragraph is only visible in **LaTeX** output.

This paragraph is hidden in HTML output but visible elsewhere.

This paragraph is visible because `show_extra` is true.

0.13 Hidden div

The hidden div executed: computed but not shown.

0.14 Raw blocks

This is raw LaTeX.

0.15 Jinja features

0.15.1 Context variables

The document title is: Kitchen Sink.

The sample size is: 50.

The current format is: latex.

0.15.2 Conditionals

Rendering to latex.

0.15.3 Loops

Features of this document:

- R code chunks
- Python code chunks
- Cross-references

0.15.4 Built-in spans

Lorem ipsum dolor sit amet consectetur adipiscing elit. Elit sed do eiusmod tempor incididunt ut labore et dolore magna.

After the page break.

0.16 Syntax highlighting

Non-executable code blocks with syntax highlighting:

```
function greet(name) {
  return ‘Hello, ${name}!‘;
}
```

```
def factorial(n: int) -> int:
    return 1 if n <= 1 else n * factorial(n - 1)
```